

ME 627: Mesh Independent Computational Techniques

Course Code	ME 627
Course Name	Mesh Independent Computational Techniques
L-T-P-C	3-0-0-3
Prerequisites	FEM in Engineering (ME 513)/Advanced FEM (ME 601)
Intended for	UG/M. Tech/MS/PhD
Distribution	Elective
Approval	recommended by the 39th BoA (Mar 2021)

Course Contents

- **Scope and Need of Mesh Independent methods**
Equations for two-dimensional and three-dimensional solids, strong-forms and weak-forms, Weighted residual method, need of mesh free methods, solution procedure of Mesh Free methods, classification according to the formulation procedures, classification according to the function approximation schemes [9 Lectures]
- **Shape function construction**
Meshfree shape function construction, point interpolation methods, moving least square shape function, interpolation error using Meshfree shape function. [6 Lectures]
- **Methods based on global weak-forms**
Meshfree radial point interpolation method and formulation, integration issues in RPI method, Imposition of essential BC in RPI method, Element Free Galerkin method formulation, and shape function, Implementation issues and application examples in EFGM, Integration techniques and enforcement of BC in EFGM. [8 Lectures]
- **Methods based on local weak-forms**
Meshless local Petrov-Galerkin method, Implementation issues and application examples. [4 Lectures]
- **eXtended Finite Element Method**
Difference between FEM and XFEM, Partition of Unity and Level set function, Intrinsic enrichment and Extrinsic enrichment, Numerical integration, Implementation and case studies [4 Lectures]
- **Computer Implementation and Case Studies**
FEM Implementation with computer programming platform
Implementation of meshfree EFGM using computer programming for analysis of (i) solid mechanics problem (ii) heat transfer problem. Enforcement of essential boundary condition will be implemented using Lagrange multiplied method.
Implementation of XFEM with extrinsic enrichment techniques for material discontinuity and crack discontinuity. [9 Lectures]

Text Books

1. G. R. Liu and Y. T. Gu, **An Introduction to meshfree Methods and their Programming**, Springer Dordrecht, Berlin, 2005.
2. Timon Rabczuk, Jeong-Hoon Song, Xiaoying Zhuang, Cosmin Anitescu, **Extended Finite Element and Mesh-free Methods**, Academic Press, 2019.

References

1. G. R. Liu, **Meshfree Methods: Moving Beyond the Finite Element Method**, CRC Press, 2018
2. **Meshfree Methods for Partial Differential Equations**, Lecture notes in Computational Science and Engineering book series, Springer, 2017.
3. Youping Chen, James Lee, Azin Eksandarin, **Meshless methods in Solid Mechanics**, Springer, 2006.
4. Hua Li, Santanu S. Mulay, **Meshless Methods and Their Numerical Properties**, CRC Press, 2013.
5. Soheil Mohammadi, **Extended finite element method for fracture analysis of Structures**, Blackwell Publishing Inc., 2008.
6. Amir R. Khoei, **Extended Finite Element Method: Theory and Applications**, John Wiley & Sons, 2014.